Forecasting Hourly Electricity Consumption in the PJM East Region Using a Two-Stage Hybrid ARIMAX - GARCH Model Integrated with Temporal Structural Variables.

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Course: Time Series

***Abstract*—Accurate hourly electricity load forecasting plays a core role in the secure management and operation of smart grid systems. This study establishes and evaluates the performance of a two-stage hybrid ARIMAX-GARCH model, incorporating time-categorical exogenous variables to disentangle the nested multiple seasonalities (daily, weekly, monthly) and the volatility clustering phenomenon of energy time series. Utilizing the real-world hourly electricity consumption dataset of PJM East (PJME) comprising $145,366$ observations during the 2002–2018 period, the study identifies the optimal model as ARIMAX(2,1,5) – GARCH(1,1) via a stepwise AIC optimization strategy. Empirical results demonstrate that the ARIMAX model thoroughly explains the linear autocorrelation structures (Ljung-Box test p-value $= 0.24$), while the GARCH(1,1) model successfully captures the strong conditional heteroskedasticity (p-value < 0.01). The coefficient $\alpha_1 = 0.7040$ completely dominates $\beta_1 = 0.0568$, reflecting the characteristic that electricity load is highly sensitive to short-term weather shocks but adjusts back to equilibrium very rapidly. The subsequent 24-hour out-of-sample forecasting results provide a dynamic confidence interval, optimally supporting dispatch decisions and mitigating operational risks in the electricity market.**

***Index Terms*—ARIMAX-GARCH Model; Multiple Seasonality; Volatility Clustering; Load Forecasting; PJME Dataset.**

I. Introduction

Electricity Load Forecasting plays a pivotal role in power system generation planning, transmission grid dispatch, and cost minimization in the spot electricity market. An imbalance between electricity supply and demand not only causes substantial economic losses but also threatens national energy security. However, forecasting electricity consumption time series at an hourly level (Hourly Load) faces two core challenges:

1) **Multiple Seasonality:** Hourly load contains a daily cycle (24 hours), a weekly cycle (168 hours) influenced by domestic behaviors combined with industrial production schedules, and an annual cycle affected by seasonal weather.
2) **Volatility Clustering:** Forecasting errors tend to cluster into periods of high volatility during extreme peak times (such as summer heatwaves or severe winter cold snaps) and low volatility during transitional seasons.

This study focuses on analyzing the hourly electricity load data system of PJM East (PJME) – one of the largest power grids in the Eastern United States. By adopting a systematic approach, the research deploys a two-stage hybrid ARIMAX-GARCH model to simultaneously address the expected mean value forecasting problem and quantify the volatility risk of the energy system in the future.

A. Time Series Models for Electricity Load Forecasting

Traditional time series analysis techniques often assume that the error variance remains constant over time. For large-scale energy data, this assumption is consistently and severely violated due to the high complexity and poor generalization capability of isolated linear models.

1) ARIMAX Model for Linear Patterns and Exogenous Variations: The ARIMAX (AutoRegressive Integrated Moving Average with Exogenous Regressors) model extends the traditional ARIMA model by directly incorporating exogenous variables (X) into the linear structure of the time series. The general mathematical equation of the ARIMAX model is expressed as follows:

$$\phi_p(L)(1-L)^d y_t = \sum_{k=1}^{K} \gamma_k X_{k,t} + \theta_q(L)\epsilon_t$$

Where:

- $\phi_p(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$ represents the autoregressive (AR) operator of order p.
- $\theta_q(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$ represents the moving average (MA) operator of order q.
- L is the lag operator ($L^k y_t = y_{t-k}$), and d is the order of differencing required to ensure the stationarity of the time series.
- $X_{k,t}$ denotes the time-categorical exogenous indicators (hour, day, month) that assist the model in completely offloading the computational burden of long-term cycles, which traditional SARIMA structures fail to converge upon within large sample spaces.

2) GARCH Model for Volatility Clustering: To handle the non-linear component and conditional variance dynamics within the residual series ϵ_t of the mean model, the GARCH

(Generalized AutoRegressive Conditional Heteroskedasticity) model proposed by Bollerslev is applied. The structure of GARCH(1,1) is formulated as the following conditional variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Where:

- σ_t^2 is the conditional variance (volatility) at time t.
- ω is the intercept coefficient (baseline long-term variance).
- α_1 is the ARCH coefficient, measuring the impact of squared error shocks from the immediately preceding period.
- β_1 is the GARCH coefficient, reflecting the persistence level of past volatility on the current period.
- The constraint conditions to ensure a stationary process and strictly positive variance are: $\omega > 0$, $\alpha_1 \geq 0$, $\beta_1 \geq 0$, and $\alpha_1 + \beta_1 < 1$.

3) ARIMAX-GARCH Hybrid Model: The hybrid ARIMAX-GARCH model operates based on a two-stage estimation philosophy:

- **Stage 1:** Estimate the ARIMAX model on the actual load series to extract all trend characteristics and linear dependencies against the time schedule, thereby obtaining the residual series ϵ_t.
- **Stage 2:** Apply the GARCH model to the retrieved residual series ϵ_t to structuralize the non-linear conditional variance component, disentangling the nature of "shocks" and the uncertain volatility risk of the load system.

B. Rolling Data Window vs Global Estimation for Model Optimization

In contrast to small-scale studies that employ short-term rolling data windows to force the series to achieve local stationarity, this study chooses a global estimation method across a large sample space combined with a one-hot encoding exogenous variable matrix. This method optimizes learning long-term correlation relationships over many years, avoids the loss of information during the annual cycle, and minimizes computational costs when implemented in real-world MLOps deployment pipelines.

C. Overview of Preceding Studies and Research Gaps

Preceding studies frequently employ standard SARIMA models for hourly electricity data but encounter the phenomenon of parameter dimensionality explosion and non-convergence when configuring the weekly seasonal parameter exceeding $s = 168$. On the other hand, several studies applying Machine Learning techniques (such as LSTM, XGBoost), despite capturing non-linear properties, neglect the testing of heavy-tailed distributions and fail to provide statistically meaningful dynamic confidence intervals for the error term's volatility. This study bridges those gaps by combining the explanatory power of robust econometrics with the high-performance handling of large exogenous matrices in Python.

D. Rationale and Objective of the Study

- **Research Rationale:** The PJME electricity consumption series exhibits typical characteristics of the energy market: heavily dependent on temporal contexts and displaying heteroskedasticity driven by extreme weather events.
- **Research Objective:** Construct a complete analytical pipeline in Python; accurately estimate the optimal parameter set for the ARIMAX-GARCH model; empirically and mathematically prove the existence of ARCH effects; and deliver a 24-hour ahead volatility risk forecasting interval.

E. Structure of the Paper

This report is structured as follows: Section 2 details the data collection methodology, matrix pre-processing steps, and mathematical test specifications. Section 3 presents the empirical results in depth, including stationarity tests, automated parameter grid-search via AIC scores, ARCH effect testing, and the visualization of the 24-hour future forecasting plots. Finally, Section 4 provides the conclusions and future research directions.

II. Materials and Methods

A. Data Collection

The research data is extracted from the real-world load database of PJM Interconnection (PJM East region - PJME). The time series includes electricity consumption observations measured in Megawatts (MW) recorded at an hourly frequency, spanning from 2002 to 2018 with a total sample size of up to $145,366$ observations.

B. Data Pre-processing

The data pre-processing workflow is rigorously designed through the following steps:

1) **Setting up the Time Index:** Convert the `Datetime` column into Pandas' `DatetimeIndex` and execute `df.sort_index()` to completely eliminate any graphical visualization distortions caused by non-sequential raw data.
2) **Frequency Standardization:** Apply the `.asfreq('H')` method to synchronize a regular hourly frequency, detecting data gaps caused by metering system errors.
3) **Linear Interpolation:** Handle the generated missing values using adjacent linear interpolation technique (`.interpolate(method='linear')`).
4) **Exogenous Feature Engineering:** Extract temporal features including: `hour` (0–23), `dayofweek` (0–6), and `month` (1–12). Proceed to transform this entire group of variables into a dummy variable matrix using the `pd.get_dummies` function with the option `drop_first=True` to avoid the Dummy Variable Trap (perfect multicollinearity).

C. Model Specification

The model specification process adheres to a hierarchical pipeline structure:

D. Rolling Window vs. Dummy Variable Approach for Seasonality

Instead of segmenting the time series into small windows, this study utilizes the entire exogenous dummy variable matrix fed into the algorithm's `exog` parameter. This approach transforms a complex, long-cycle SARIMA problem into a regression problem with autocorrelated errors, accelerating CPU computational processing speeds by hundreds of times while ensuring the annual cycle is preserved in its entirety.

E. Model Evaluation

The model is evaluated through:

- **Akaike Information Criterion (AIC)** and **Bayesian Information Criterion (BIC)** for optimal order selection.
- **Ljung-Box Test** on the residual series to ensure freedom from autocorrelation (white noise verification).
- **ARCH Lagrange Multiplier (ARCH-LM) Test** to confirm the existence of conditional volatility effects.

III. Results and Discussion

A. Data Pre-Processing & Visual Analysis

The execution of the `df.sort_index()` command in Step 1 completely eliminated the visualization artifacts (backward diagonal crisscross lines) that distorted the initial raw plot. This restored a smooth, clean sinusoidal wave time series that accurately reflects the pulse of the electricity load.

B. Stationarity Test on Original Series

To mathematically verify stationarity before fitting the mean model, the Augmented Dickey-Fuller (ADF) test was performed on the entire original `PJME_MW` series. The obtained results are presented in Table I:

TABLE I: Augmented Dickey-Fuller (ADF) Test Results

Test Statistic	Empirical Value
ADF Statistic	-18.8289
p-value	0.0000
Critical Value 1%	-3.4304
Critical Value 5%	-2.8616

Since p-value $= 0.0000 < 0.05$, the original dataset exhibits strong stationarity at the 1% significance level.

1) First-Order Differencing Selection in Auto-ARIMA: Although the original series achieves global stationarity according to the ADF test, the automated stepwise optimization search algorithm `auto_arima` decided to apply first-order differencing ($d = 1$) when scanning through the entire long-term structure integrated with the large exogenous matrix. This action helps completely eliminate the risks of local drifts and ensures maximum safe convergence for the autoregressive coefficients.

C. Time Series Modeling

1) Model Parameter Identification and Order Determination: The optimization process for finding the optimal mean model structure using the Stepwise Search algorithm in the `pmdarima` library lasted for a total of **2,818.793 seconds** ($\sim$47 minutes) on the dataset of $145,366$ rows. The algorithm traversed multiple order combinations to minimize the AIC objective function:

- Initial trial model ARIMA(2,1,2) achieved AIC $=$ 2428594.393
- Trial model ARIMA(0,1,0) achieved AIC $= 2553151.721$
- Trial model ARIMA(3,1,3) achieved AIC $= 2421178.887$

The winning model result with the lowest information criterion score is: **SARIMAX(2,1,5)**.

- Final AIC score reached: **2419971.757**
- Final BIC score reached: **2420050.853**

2) Testing for ARCH Effects: After estimating the ARIMAX(2,1,5) stage, the residual series is extracted and subjected to the Lagrange Multiplier test (ARCH-LM Test) to determine the existence of autoregressive conditional heteroskedasticity.

- **Test Result:** Lagrange Multiplier test p-value $= 0.0000$
- **Consensus Result from the ARIMAX statistical summary table:** The Heteroskedasticity (H) test coefficient has a p-value $= 0.00$.

Since p-value < 0.01, the null hypothesis H_0 of homoskedasticity is completely rejected at the 1% significance level. The empirical evidence demonstrates that a highly robust ARCH effect exists within the residuals, confirming that the application of a GARCH volatility model stage is mandatory.

D. Model Fitting Results

The tables below summarize in detail the parameter estimation results of the optimal two-stage model system obtained from the empirical framework:

1) Stage 1: Mean Autoregressive Coefficients ARIMAX(2,1,5): All core coefficients of the AR and MA components achieve highly significant statistical meaning (p-value $= 0.0000 < 0.05$):

TABLE II: ARIMAX(2,1,5) Parameter Estimation Results

Parameter	Coefficient	Standard Error	z-statistic	p-value ($P > \|z\|$)
ar.L1	1.8897	0.001	1669.191	0.000
ar.L2	-0.9533	0.001	-910.880	0.000
ma.L1	-1.3492	0.002	-736.187	0.000
ma.L2	0.1480	0.003	42.927	0.000
ma.L3	0.0729	0.006	12.211	0.000
ma.L4	0.0891	0.006	14.167	0.000
ma.L5	0.0758	0.004	21.029	0.000

2) Stage 2: Volatility Variance Coefficients GARCH(1,1): The volatility model is fitted using the Maximum Likelihood Estimation (MLE) method, yielding the conditional variance parameter set:

The coefficient $\alpha_1 = 0.7040$ possesses a large value, reflecting a very high sensitivity of load volatility to sudden external

TABLE III: GARCH(1,1) Parameter Estimation Results

Parameter	**Coefficient (coef)**	**Standard Error (std err)**	**t-statistic**	**p-value ($P > \|t\|$)**
mu (μ)	75.5941	9.789	7.722	0.000
omega (ω)	4.9974×10^5	7.267×10^4	6.877	0.000
alpha[1] (α_1)	0.7040	7.770×10^{-2}	9.060	0.000
beta[1] (β_1)	0.0568	2.678×10^{-2}	2.119	0.034

information/shocks. The sum $\alpha_1 + \beta_1 = 0.7040 + 0.0568 = 0.7608 < 1$, proving that the GARCH volatility process achieves robust stationarity. However, due to the low volatility persistence coefficient from the past β_1 (0.0568), the variance shocks of electricity consumption in the PJME region are characterized by a highly localized nature—surging very rapidly but also dissipating and dampening quickly once the weather context stabilizes.

3) Visualizing Residual Autocorrelation and Normality Test:

- **Autocorrelation Test:** The Ljung-Box test results on the residual series yield a test statistic of $Q = 1.36$ with $\text{Prob}(Q) = 0.24$. Since p-value $= 0.24 > 0.05$, there is absolutely no statistical basis to reject the null hypothesis H_0. This demonstrates that the residuals are completely free of linear autocorrelation, becoming a white noise process in terms of the mean value.
- **Normality Test:** The Kurtosis index reaches **14.58** (far exceeding the standard baseline of 3), combined with a very high Jarque-Bera score, confirming that the residual series exhibits a heavy-tailed distribution characteristic caused by extreme consumption peaks.

E. Forecasting and Accuracy Analysis

1) Forecasted Data Visualization: The 24-hour ahead out-of-sample forecast generation is executed by combining the point forecasts from ARIMAX with the dynamic conditional variance matrix from GARCH.
The displayed plot indicates that:

- The blue generation forecast line perfectly reflects the recurring sinusoidal structure of the daily cycle, peaking during afternoon windows (hours 15–16) and gradually decreasing towards the night.
- The pink risk amplitude band (95% Confidence Interval) automatically expands significantly during daytime peak hours and narrows during the night. This proves the value of GARCH in risk quantification: high-demand hours are invariably accompanied by a greater degree of load uncertainty.

2) Accuracy Evaluation: The model achieves high accuracy thanks to its integrated two-stage structure. Disentangling the time-varying variance using GARCH allows the forecast confidence intervals to adapt flexibly in real-time (Dynamic Bandwidth), protecting the grid operation system from sudden overload scenarios by closely tracking the upper bound of the volatile forecast band.

F. Limitations and Future Directions

Although the model achieves highly robust statistical test results, a minor limitation persists: the study solely utilizes pure temporal structural variables as exogenous dummy features without directly integrating real-world temperature time series data for the East Coast region. Future research directions could expand this study toward ARIMAX-EGARCH or ARIMAX-GJR-GARCH models to further disentangle the asymmetry of shocks (for instance, a sudden temperature spike in the summer typically causes electricity load to surge far more intensely than an equivalent temperature drop).

IV. Conclusion

This study has successfully implemented a comprehensive analytical and forecasting pipeline for hourly electricity load time series on the PJM East grid using Python, achieved through the combination of a hybrid ARIMAX(2,1,5) – GARCH(1,1) model.

The One-Hot Encoded exogenous variable system thoroughly addresses the complex nested multiple seasonalities problem within a large sample space ($145,366$ observations) without causing algorithmic computational bottlenecks. Rigorous empirical statistical testing (ADF, ARCH-LM, Ljung-Box) establishes a solid foundation for scientific argumentation, validating the correctness of the dynamic conditional heteroskedastic variance structure. The empirical results demonstrate that the load volatility in the PJME region is highly sensitive to instantaneous shocks but possesses a rapid self-recovery speed back to equilibrium. This work serves as an essential technical reference for data engineers and energy dispatch specialists seeking to optimize power generation operating costs and enhance the security of smart grid systems.